

Scenario:

You're working on a personal project named "<YourLastname>_local_project" and want to experiment with new features without affecting the main codebase.

Example: *Mansueto_local_project*

 Lita_local_project 10/3/2025 2:21 PM File folder

Note:

- Ensure Git is installed on your machine.
- Set up your Git username and email (if not already configured)

```
PS C:\Users\ACER\Desktop\Lita_local_project> git config --global user.name "Keith"
PS C:\Users\ACER\Desktop\Lita_local_project> git config --global user.email "clita.a12241743@umak.edu.ph"
```

- Create a directory called <YourLastname>_local_project, navigate into it, and initialize it as a Git repository

```
PS C:\Users\ACER\Desktop\Lita_local_project> mkdir Lita_local_project
```

- Multiple commits reflecting changes in different branches.
- Evidence of conflict resolution during branch merging (optional).
- A Git tag-marking version v1.0 (optional).

Tasks:**1. Create a new branch:**

- Create a new branch named "**feature_x**" to isolate your experimental changes.

```
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git checkout -b feature_x
Switched to a new branch 'feature_x'
```

2. Stage changes:

- Make modifications to your project files.
- Stage the changes using the git add command.

```
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> echo "This is feature_x work" > feature.txt
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git add feature.txt
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git commit -m "Added feature_x experimental file"
[feature_x 9448517] Added feature_x experimental file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 feature.txt
```

3. Commit changes:

- Commit the staged changes to your local repository using the git commit command. Provide a clear and concise commit message.

```
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> echo "More updates for feature_x" >> feature.txt
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git add feature.txt
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git commit -m "Updated feature_x with additional changes"
```

4. Switch back to the main branch:

- Use the git checkout command to switch back to the main branch.

```
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git checkout master
Switched to branch 'master'
```

```
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> echo "Main branch content" > main.txt
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git add main.txt
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git commit -m "Added main branch file"
[feature_x a7cc5ac] Added main branch file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 main.txt
```

5. Merge changes from the feature branch:

- Use the git merge command to merge the changes from the "feature_x" branch into the main branch. Resolve any merge conflicts that may arise.

```
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git merge feature_x
Already up to date.
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git add .
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git commit -m "Resolved merge conflict between main and feature_x"
On branch feature_x
nothing to commit, working tree clean
```

6. Viewing the Commit History:

- Use Git commands to view the commit history and ensure that all commits are in place.

```
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git log --oneline --graph --all
* a7cc5ac (HEAD -> feature_x) Added main branch file
* 4becf22 Updated feature_x with additional changes
* 9448517 Added feature_x experimental file
* 3460e26 (master) Initial commit: project setup
```

7. Tagging Versions (Optional):

- After completing the merge, use the git tag command to create a tag named v1.0 to mark the first version of your project.

```
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git log --oneline --graph --all
* a7cc5ac (HEAD -> feature_x) Added main branch file
* 4becf22 Updated feature_x with additional changes
* 9448517 Added feature_x experimental file
* 3460e26 (master) Initial commit: project setup
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git tag v1.0
PS C:\Users\ACER\Desktop\Lita_local_project\Lita_local_project> git log --graph
* commit a7cc5ac6addab72320c67ab8e1b2438fd5fb81c8 (HEAD -> feature_x, tag: v1.0)
| Author: Keith <clita.a12241743@umak.edu.ph>
| Date:   Fri Oct 3 14:21:42 2025 +0800
|
|     Added main branch file
|
| * commit 4becf224fb66a7c9788c0b3bf14ca21fc40f10f2
| | Author: Keith <clita.a12241743@umak.edu.ph>
| | Date:   Fri Oct 3 14:21:18 2025 +0800
| |
| |     Updated feature_x with additional changes
| |
| | * commit 944851751dff706055fe6d250c7553f806124e03
| | | Author: Keith <clita.a12241743@umak.edu.ph>
| | | Date:   Fri Oct 3 14:21:04 2025 +0800
| | |
| | |     Added feature_x experimental file
| | |
| | * commit 3460e269e572930c86aca01233d20264262aa3a1 (master)
| | | Author: Keith <clita.a12241743@umak.edu.ph>
| | | Date:   Fri Oct 3 14:20:41 2025 +0800
| | |
| | |     Initial commit: project setup
```